THE CAP ACTION ON THE CYCLE CLASS OF A FLAT COHERENT SHEAF AGAINST A COHERENT CYCLE CLASS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The family of r-cycles associated to a coherent sheaf F on X with fibre supports of dimension at most r (relative-cycles chapter, Example example-family-associated-module, canonical tag 0H4Q). cited at statement lines 2365-2366. Defines alpha = [F/X/Y]_r appearing in the statement.
- The cycle associated to a coherent sheaf of delta-dimension at most e on Y (Chow Homology, Definition chow-definition-cycle-associated-to-coherent-sheaf, canonical tag 02QX). cited at statement lines 2367-2368. Defines beta = [G]_e appearing in the statement.

Source reference. Original reference 0H4Q: sources/stacks-chow/tags.txt, source lines 21360–21360. Cited at statement lines 2365-2366: defines alpha = [F/X/Y]_r.

Source reference. Original definition 02QX: sources/stacks-chow/tags.txt, source lines 3604–3604. Cited at statement lines 2367-2368: defines beta = [G]_e.

Lemma 0.2. *Let (S, δ) and $f : X \to Y$ be as above. Let $\mathcal{F}$ be a coherent $\mathcal{O}_X$-module with $\dim(\mathit{Supp}(\mathcal{F}_y)) \leq r$ for all $y \in Y$. Let $\mathcal{G}$ be a coherent $\mathcal{O}_Y$-module with $\dim_\delta(\mathit{Supp}(\mathcal{G})) \leq e$. Set $\alpha = [\mathcal{F}/X/Y]_r$ (Example 0H4Q) and $\beta = [\mathcal{G}]_e$ (Chow Homology, Definition 02QX). If $\mathcal{F}$ is flat over Y, then $\alpha \cap \beta = [\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}]_{r+e}$.*

Proof. Observe that

$$\mathrm{Supp}(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) = \mathrm{Supp}(\mathcal{F}) \cap f^{-1}\mathrm{Supp}(\mathcal{G}) = \bigcup\nolimits_{y \in \mathrm{Supp}(\mathcal{G})} \mathrm{Supp}(\mathcal{F}_y)$$

It follows that this is a closed subset of δ-dimension $\leq r + e$. Whence the expression $[\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}]_{r+e}$ makes sense.

We will use the notation $\beta = \sum n_i[Z_i]$, $y_i \in Z_i$, $\alpha_{y_i} = \sum m_{ij}[V_{ij}]$, and $\overline{V}_{ij}$ introduced in the construction of $\alpha \cap \beta$. Since $\beta = [\mathcal{G}]_e$ we see that the Z_i are the irreducible components of $\mathrm{Supp}(\mathcal{G})$ which have δ-dimension e. Similarly, the V_{ij} are the irreducible components of $\mathrm{Supp}(\mathcal{F}_{y_i})$ having dimension r. It follows from this and the equation in the first paragraph that $\overline{V}_{ij}$ are the irreducible components

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of $\mathrm{Supp}(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})$ having δ-dimension $r + e$. Thus to prove the lemma it now suffices to show that

$$\mathrm{length}_{\mathcal{O}_{X,\xi_{ij}}}((\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})_{\xi_{ij}}) = \mathrm{length}_{\mathcal{O}_{X_{y_i},\xi_{ij}}}((\mathcal{F}_{y_i})_{\xi_{ij}}) \cdot \mathrm{length}_{\mathcal{O}_{Y,y_i}}(\mathcal{G}_{y_i})$$

By the first paragraph of the proof the left hand side is equal to the length of the $B = \mathcal{O}_{X,\xi_{ij}}$-module

$$\mathcal{G}_{y_i} \otimes_{\mathcal{O}_{Y,y_i}} \mathcal{F}_{\xi_{ij}} = M \otimes_A N$$

Here $M = \mathcal{G}_{y_i}$ is a finite length $A = \mathcal{O}_{Y,y_i}$-module and $N = \mathcal{F}_{\xi_{ij}}$ is a finite B-module such that $N/\mathfrak{m}_A N$ has finite length. Since $\mathcal{F}$ is flat over Y the module N is A-flat. The right hand side of the formula is equal to

$$\mathrm{length}_B(N/\mathfrak{m}_A N) \cdot \mathrm{length}_A(M)$$

Thus the right and left hand side of the formula are additive in M (use flatness of N over A). Thus it suffices to prove the formula with $M = \kappa_A$ is the residue field in which case it is immediate. $\square$